

Curriculum Vitae: Jennet Baumbach, Ph.D

✉ jennet.baumbach@mail.utoronto.ca

📁 Portfolio

🎓 Google Scholar

🌐 LinkedIn

University Teaching Experience

- 2023 – Present 📖 **Sessional Lecturer I** | University of Toronto 📍 Mississauga & Toronto, ON.
- Serve as instructor for undergraduate and graduate courses.
 - Design and deliver original course content.
 - Lead and coordinate teams of teaching assistants.
- 2022 – Present 📖 **Tutorial Leader** | University of Toronto 📍 Mississauga & Toronto, ON.
- Lead graduate and undergraduate tutorials focused on statistical reasoning and problem-solving.
 - Teach the mathematical and theoretical foundations of statistical modeling.
 - Train students to interpret, critique, and clearly communicate statistical analyses.
- 2019 – 2025 📖 **Writing TA** | University of Toronto 📍 Mississauga, ON.
- Provided detailed feedback on a two-part scaffolded assignment about critical thinking in science communication (Psy290 - Intro to Neuroscience).
- 2019 – Present 📖 **Grading TA** | University of Toronto 📍 Mississauga & Toronto, ON.
- Evaluated papers and tests for large undergraduate classes under strict academic standards and deadlines.
- 2017 – 2019 📖 **Lab Demonstrator** | Brock University 📍 St. Catharines, ON.
- Instructed undergraduate laboratory courses involving physiological measurement, neuroanatomical dissections, and data collection.
 - Taught statistical analysis and data visualization techniques to interpret experimental results.
- 📖 **Seminar Leader** | Brock University 📍 St. Catharines, ON.
- Facilitated discussion-based seminars emphasizing evidence-based critical thinking and respectful debate.

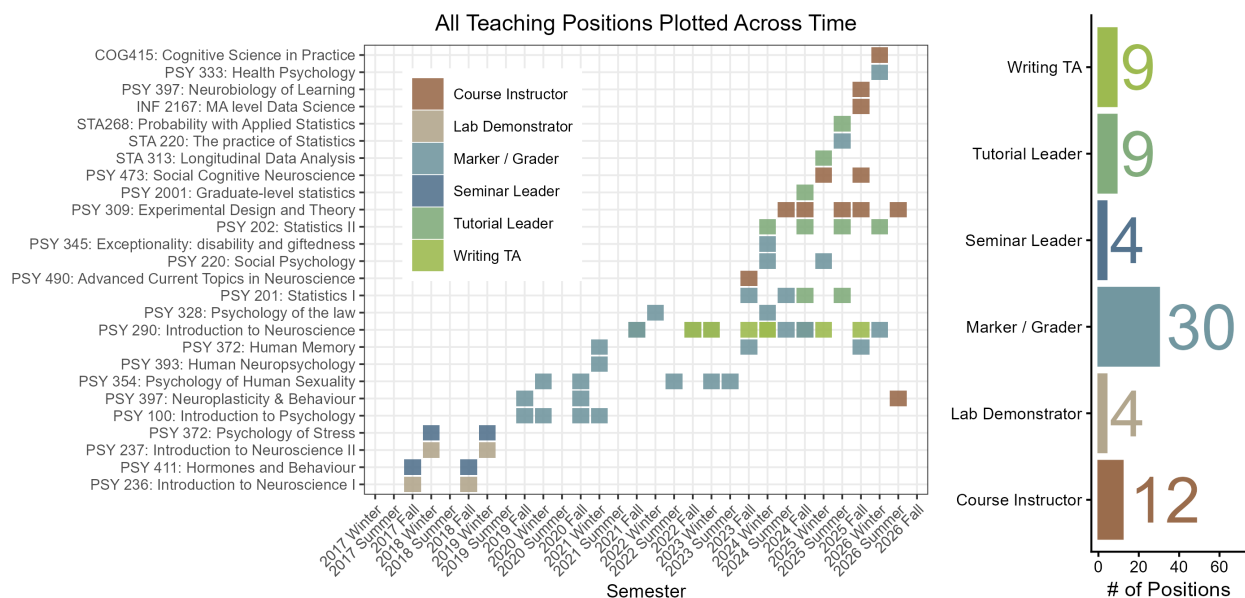


Figure 1. All teaching positions plotted across time and tallied, coloured by teaching role.

Example Teaching Materials

- 1  **INF2167 - R for Data Science (2025)** |  [LINK](#)
 - Hosted in-class sessions where students followed along as I wrote, explained, and debugged code in real time.
- 2  **Psy309 - Computational Project in Research Methods and Statistics (2024)** |  [LINK](#)
 - Designed a reproducible computational project for a third-year research methods course, guiding students through statistical analysis and data visualization using simulated data.
- 3  **Statistics Review Materials - University of Toronto Department of Statistics (2025)** |  [LINK](#)
 - Developed interactive, code-based learning materials to support conceptual and mathematical understanding of core statistical topics (e.g., covariance and correlation).

Education

- 2019 – 2025  **Ph.D Psychology**, University of Toronto |  [Link to PhD Thesis](#)
- 2017 – 2019  **MA Psychology**, Brock University |  [Link to MA Thesis](#)
- 2012 – 2017  **hBA Psychology**, Brock University

Professional Work Experience

- 2024 – Present  **Biostatistician** | St. Josephs Health Care Centre  London, ON.
- Lead the design and implementation of quantitative and qualitative research studies in health sciences.
 - Analyze large administrative datasets using advanced statistical & data science methods.
 - Develop publication-quality data visualizations and lead manuscript preparation for peer-reviewed journal submissions.
-  **Research Assistant** | University of Toronto  Toronto, ON.
- Develop instructional content and video materials linking statistical theory to computational data analysis.
 - Collaborate with faculty to enhance undergraduate statistics instruction through applied examples
 - Create data visualizations, interactive tools, and reproducible analysis scripts to support student learning
- 2025 – Present  **Research Consultant** | Self-Employment
- Work with faculty members to refine grant proposals, analyze data, and prepare manuscripts for peer-review.
-  **Invigilator - Accessibility Services** | University of Toronto  Mississauga, ON.
- Administer and oversee tests and exams in the accomodated testing center in accordadnce with institutional policies and accessibility standards.

Competative Funding

2023 – 2025	 Women in Science and Mathematics Fellowship \$11,000; Awarded on behalf of General Motors.
2022 – 2023	 Pain Scientist Award \$12,500; Awarded on behalf of the University of Toronto Center for the Study of Pain.
2020 – 2023	 NSERC Post-Graduate Fellowship \$63,000; Awarded on behalf of the Canadian Government.
2022	 Canadian Pain Society - Best Poster Presentation \$150; Awarded at the CPS Annual meeting - Selected from >250 posters.
2019 – 2023	 Faculty of Arts and Science Top Doctoral Fellowship \$92,000; Awarded on behalf of the University of Toronto Department of Arts & Sciences.
2019 – 2020	 Domestic Graduate Acceptance Award \$2,500; Awarded on behalf of the University of Toronto Faculty of Graduate Studies.
2018 – 2019	 Ontario Graduate Scholarship \$15,000; Awarded on behalf of the Government of Ontario.
2017 – 2018	 Dean of Graduate Studies Award \$4,500; Awarded on behalf of Brock University Faculty of Graduate Studies.
2015 – 2016	 Match of Minds Fellowship \$11,000; Awarded on behalf of the Psychology Department of Brock University

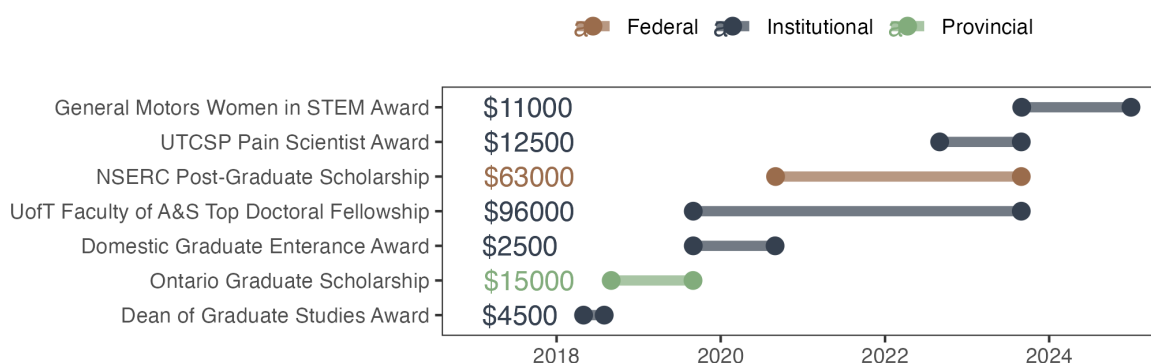


Figure 2. Visual depiction of graduate school funding won over time.

Technical Skills

Research Design	 Human and animal research design; protocol development; ethics submissions; longitudinal and experimental study design; program-level research planning.
Survey Generation	 Survey design, validation, and deployment using Qualtrics, Google Forms, and REDCap.
Data Analysis	 General linear models; multilevel and mixed-effects models; structural equation modeling; Bayesian analysis; A/B hypothesis testing; qualitative methods; advanced data visualization.

Technical Skills (continued)

Coding	 R (tidyverse, RMarkdown, Shiny); SPSS; Excel. Reproducible workflows; dynamic document generation; LaTeX-based typesetting; version-controlled, readable, and modular code.
Scientific Communication	 Peer-reviewed manuscript preparation; conference posters and presentations; code-based figure and presentation generation; communication of technical results to academic, clinical, and non-technical audiences.

Research Contributions

Peer-Reviewed Journal Articles (DOIs are linked)

- 1 **J. L. Baumbach**, C. Y. Y. Mui, A. M. Leonetti, and L. J. Martin, "A history of injury enhances affective and sensory responses to predator threat by sensitizing corticosterone release through trpa1 receptor signaling," *Current Biology*, 2025.  DOI: 10.1016/j.cub.2025.07.005.
- 2 **J. L. Baumbach**, C. Y. Y. Mui, A. M. Leonetti, and L. J. Martin, "Corticosterone regulates the balance between freezing and rearing in defensive responses to predator threat," *Progress in Neuro-Psychopharmacology and Biological Psychiatry*, 2025.  DOI: 10.1016/j.pnpbp.2025.111579.
- 3 **J. L. Baumbach**, A. M. Leonetti, and L. J. Martin, "Inflammatory injury induces pain sensitization that is expressed beyond the site of injury in male (and not in female) mice," *Behavioural Brain Research*, vol. 475, p. 115 215, Oct. 2024, ISSN: 01664328.  DOI: 10.1016/j.bbr.2024.115215.
- 4 **J. L. Baumbach**, C. Y. Mui, F. Tuz Zahra, and L. J. Martin, "A single exposure to the predator odor 2,4,5-trimethylthiazoline causes long-lasting affective behavioral changes in female mice: Modulation by kappa opioid receptor signaling," *Pharmacology Biochemistry and Behavior*, vol. 242, p. 173 822, Sep. 2024, ISSN: 00913057.  DOI: 10.1016/j.pbb.2024.173822. (visited on 09/04/2024).
- 5 S. Darvish-Ghane, **J. Baumbach**, and L. J. Martin, "Influence of Inflammatory Pain and Dopamine on Synaptic Transmission in the Mouse ACC," en, *International Journal of Molecular Sciences*, vol. 24, no. 13, p. 11 113, Jul. 2023, ISSN: 1422-0067.  DOI: 10.3390/ijms241311113. (visited on 04/15/2024).
- 6 R. A. Herlehy, S. Lim, S. H. Murray, **J. L. Baumbach**, M. Van Wingerden, and C. M. McCormick, "Effect of social instability stress in adolescence or adulthood on sensitivity to sucrose concentration in a social context in male and female Long-Evans rats," en, *Developmental Psychobiology*, vol. 64, no. 6, e22293, Sep. 2022, ISSN: 0012-1630, 1098-2302.  DOI: 10.1002/dev.22293. (visited on 04/15/2024).
- 7 **J. Baumbach** and C. McCormick, "Nicotine sensitization (part 1): Estradiol or tamoxifen is required during the induction phase and not the expression phase to enable locomotor sensitization to nicotine in female rats," *Psychopharmacology*, vol. 238, no. 2, 2021, ISSN: 14322072.  DOI: 10.1007/s00213-020-05685-8.
- 8 **J. Baumbach** and C. McCormick, "Nicotine sensitization (Part 2): Time spent in the centre of an open field sensitizes to repeated nicotine into the drug-free state in female rats," *Psychopharmacology*, vol. 238, no. 2, 2021, ISSN: 14322072.  DOI: 10.1007/s00213-020-05686-7.
- 9 G. Stefanelli, C. Makowski, M. Brimble, *et al.*, "The histone chaperone Anp32e regulates memory formation, transcription, and dendritic morphology by regulating steady-state H2A.Z binding in neurons," *Cell Reports*, vol. 36, no. 7, 2021, ISSN: 22111247.  DOI: 10.1016/j.celrep.2021.109551.
- 10 **J. L. Baumbach** and I. B. Zovkic, "Hormone-epigenome interactions in behavioural regulation," *Hormones and Behavior*, vol. 118, no. November 2019, p. 104 680, 2020, Publisher: Elsevier, ISSN: 10956867.  DOI: 10.1016/j.yhbeh.2020.104680.
- 11 M. Marcolin, **J. Baumbach**, T. Hodges, and C. McCormick, "The effects of social instability stress and subsequent ethanol consumption in adolescence on brain and behavioral development in male rats," *Alcohol*, vol. 82, 2020, ISSN: 18736823.  DOI: 10.1016/j.alcohol.2019.08.003.

- 12 C. McCormick, K. Smith, **J. Baumbach**, *et al.*, "Adolescent social instability stress leads to immediate and lasting sex-specific changes in the neuroendocrine-immune-gut axis in rats," *Hormones and Behavior*, vol. 126, 2020, ISSN: 10956867.  DOI: 10.1016/j.yhbeh.2020.104845.
- 13 F. Ramzan, **J. Baumbach**, A. D. Monks, and I. B. Zovkic, "Histone H2A.Z is required for androgen receptor-mediated effects on fear memory," *Neurobiology of Learning and Memory*, vol. 175, no. September, p. 107311, 2020, Publisher: Elsevier, ISSN: 10959564.  DOI: 10.1016/j.nlm.2020.107311.
- 14 F. Ramzan, S. D. Creighton, M. Hall, *et al.*, "Sex-specific effects of the histone variant H2A.Z on fear memory, stress-enhanced fear learning and hypersensitivity to pain," *Scientific Reports*, vol. 10, no. 1, pp. 1–17, 2020, Publisher: Nature Publishing Group UK ISBN: 0123456789, ISSN: 20452322.  DOI: 10.1038/s41598-020-71229-x.
- 15 J. Simone, **J. Baumbach**, J. McPherson, and C. McCormick, "Adolescent CB1 receptor antagonism influences subsequent social interactions and neural activity in female rats," *International Journal of Developmental Neuroscience*, vol. 80, no. 4, 2020, ISSN: 1873474X.  DOI: 10.1002/jdn.10028.
- 16 M. Marcolin, T. Hodges, **J. Baumbach**, and C. McCormick, "Adolescent social stress and social context influence the intake of ethanol and sucrose in male rats soon and long after the stress exposures," *Developmental Psychobiology*, vol. 61, no. 1, 2019, ISSN: 10982302.  DOI: 10.1002/dev.21800.
- 17 T. Hodges, **J. Baumbach**, and C. McCormick, "Predictors of social instability stress effects on social interaction and anxiety in adolescent male rats," *Developmental Psychobiology*, vol. 60, no. 6, 2018, ISSN: 10982302.  DOI: 10.1002/dev.21626.
- 18 J. Simone, **J. Baumbach**, and C. McCormick, "Effects of CB1 receptor antagonism and stress exposures in adolescence on socioemotional behaviours, neuroendocrine stress responses, and expression of relevant proteins in the hippocampus and prefrontal cortex in rats," *Neuropharmacology*, vol. 128, 2018, ISSN: 18737064.  DOI: 10.1016/j.neuropharm.2017.10.029.
- 19 J. Simone, **J. Baumbach**, and C. McCormick, "Sex-specific effects of CB1 receptor antagonism and stress in adolescence on anxiety, corticosterone concentrations, and contextual fear in adulthood in rats," *International Journal of Developmental Neuroscience*, vol. 69, 2018, ISSN: 1873474X.  DOI: 10.1016/j.ijdevneu.2018.07.011.
- 20 T. Hodges, **J. Baumbach**, M. Marcolin, R. Bredewold, A. Veenema, and C. McCormick, "Social instability stress in adolescent male rats reduces social interaction and social recognition performance and increases oxytocin receptor binding," *Neuroscience*, vol. 359, 2017, ISSN: 18737544.  DOI: 10.1016/j.neuroscience.2017.07.032.

Select Academic Presentations

- | | |
|------|--|
| 2026 |  McIntyre, A. M., Baumbach, J. L. , & Lin, G. Symposium: "Exploring the Prevalence and Impact of Post-Stroke Spasticity in Long-Term Care" 37th International Nursing Research Congress 
Toronto, ON |
| |  Baumbach, J. L. , Booth, R., Forchuk, C., & McIntyre, A. M. "From Triage to Admission: Evaluating Acuity Assignment and Outcomes in Emergency Department Stroke Care" Annual Meeting of the Emergency Nurses Association.  Phoenix, AZ |
| 2022 |  Baumbach, J. L. & Martin, L. J. "Pain Facilitates Chronification of Predator Stress-Induced Mechanical Hypersensitivity in Male Mice". Annual Meeting of the Canadian Pain Society 
Montreal, QC
<i>Won first place in the poster competition out of over 400 posters.</i> |

Example Course Evaluations

7. Please comment on the overall quality of the instruction in this course.

Comments
The instruction was clear and Jennet greatly encouraged student participation. The assignments and her lectures encouraged us to think more deeply about the difference between results and author interpretations when engaging with journal articles. I think this is a very important skill and it is good that she focused on it.
Jennet is an incredible course instructor. She takes the time to explain every intricacy, so that the entire class can understand the complex subjects we are discussing.
One of the best instruction I have ever gotten. UTM should look into more seminar courses as this course was very intellectually stimulating and helped me get a better understanding for the field of neuroscience. Courses like this encourage thinking and ideas!!! Super glad I took this course and would highly recommend to anyone! I really hope more students get to have an experience like this with an instructor as good as Jennet! I genuinely look at scientific articles in a more critical way now and feel like I have grown as a student because of this course! My instructor was a 5 star 10/10 and deserves tenure !!
Professor Jennet made the course incredibly interesting and fun to attend; the structure of how she conducted it, how she encouraged and facilitated discussion in the class, and provided help in office hours and breaks during lecture was incredibly helpful.
I think Jennet is a great instructor and I think this is because of how passionate she is about neuroscience. It's always so endearing to be taught by someone who loves what they're doing. I really wish we had more of these seminar/discussion-based courses. They are so much more engaging and they allow students to really express what they've learned. I think I would've really enjoyed more of my university life with more seminar classes lol. I just get so socially anxious participating in a giant lecture hall, y'know?
Very interesting lectures, amazing quality course where I learned a lot of useful knowledge.
I found the format of the course to have been very captivating. It was unique to all other class formats taken in my undergrad and I felt it provide a great opportunity for me to become comfortable with having thoughtful discussions with my peers and driving me to construct more meaningful questions on different subjects.
The course instructor provided a very welcoming environment. Speaking up in seminars when the class size is small is challenging for me, but her warm and nurturing demeanor made it easier. She was also very flexible with her hours to help students out outside class and office hours which was extremely helpful. In addition to being helpful with course materials and presentations, she provided additional help with graduate applications and offers good advice that i may not have necessarily understood.

8. Please comment on any assistance that was available to support your learning in this course.

Comments
There were office hours and Jennet encouraged questions that needed clarification before moving on with class content during lectures. She also provided links to articles that spoke more about concepts covered in class just in case people were interested to learn more
Any time you need help with course content, Jennet was available and more than willing to explain. Additionally, Jennet coached us through many other subjects that were stressful in our general scholastic life, such as grad school applications.
My professor was super enthusiastic to help in office hours and through email with anything. She always created a safe learning space and encouraged discussion!
Professor Jennet was available 24/7 over email and was incredibly welcoming during office hours and encouraged reaching out whenever we felt stuck or had any questions.
Office hours, email.
Answered all questions very clearly and in-depth!
Jennett provided a variety of material that was not only conducive to success in the course material but also very useful for further endeavours in undergrad and graduate work.
She was very flexible and helpful when I came to her for help regarding course papers, assignments, and test help.